IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2014-15

EEE/EIE PART III/IV: MEng, BEng and ACGI

Corrected Copy

CONTROL ENGINEERING

Friday, 19 December 9:00 am

Time allowed: 3:00 hours

9:35Am
CORRECTION ON
QUESTION 2 a.ii)
SEE INSIDE

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : A. Astolfi
Second Marker(s) : D. Angeli

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CONTROL ENGINEERING

1. Consider a linear, single-input, discrete-time, system of dimension $n = 3$, that is $x = [x_1, x_2, x_3]'$, with

$$A = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

a) Study the reachability and controllability properties of the system. [4 marks]

b) Compute the set of states $\mathscr{R}_2$ that can be reached from $x(0) = 0$ in two steps. [4 marks]

c) Determine the set of states $\mathscr{C}_2$ that can be controlled to zero in two steps. Explain why the set $\mathscr{C}_2$ is larger than the set $\mathscr{R}_2$. [6 marks]

d) Consider a state feedback control law $u = Kx$ and determine the set of all matrices K such that the matrix $A + BK$ has all eigenvalues equal to 0. Explain why the matrix assigning the closed-loop eigenvalues exists, but it is not unique, and determine the matrix K which is such that KK^T is minimal. [6 marks]

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2. Consider the problem of regulating the temperature of a shower. The system can be modeled as a linear discrete-time system described by the equation

$$T(t+h) = u(t),$$

in which h is a positive integer, T denotes the temperature of the water, and u is the user selected control signal. The constant h models how fast the boiler is to react to the user command. Most people update u depending on the difference between the current temperature and their favourite temperature T_0, thus implementing the equation

$$u(t) = u(t-1) - \alpha(T(t) - T_0),$$

where $\alpha \in (1,2)$.

a) Assume $h = 1$, that is the boiler is *fast*.

i) By eliminating the variable u write the equation of the closed-loop system as a difference equation in the variable $T(t)$ with input T_0. [2 marks]

ii) Study the stability properties of the system determined in part a.i) as a function of the parameter $\alpha \in (1,2)$. [2 marks]

b) Assume $h = 2$, that is the boiler is *slow*.

i) By eliminating the variable u write the equation of the closed-loop system as a difference equation in the variable $T(t)$ with input T_0. [4 marks]

ii) Let $T(t)$ be the output of the system. Determine a state space representation for the system, that is write the equation of the system in the form

$$x^+ = Ax + Bu, \qquad y = Cx,$$

where x is the state vector with two components, $y(t) = T(t)$ and $u(t) = T_0$. Write explicitly the matrices A, B and C. [6 marks]

iii) Study the stability properties of the system determined in part c.ii) as a function of the parameter $\alpha \in (1,2)$. [4 marks]

c) Explain why a fast boiler gives a better shower. [2 marks]

3. A linear, continuous-time, descriptor system is a system described by the equations

$$E\dot{x} = Ax + Bu, \qquad y = Cx,$$

in which $x(t) \in \mathbb{R}^n$ denotes the state of the system, $u(t) \in \mathbb{R}^m$ the input signal and $y(t) \in \mathbb{R}^p$ the output signal. The matrix $E \in \mathbb{R}^{n \times n}$ is singular, whereas the matrices A, B and C are defined as for standard linear systems.

Descriptor systems often arise as the result of a modeling process whenever too many states have been introduced and can be transformed, under certain conditions, into standard linear systems using the simple procedure described in what follows for a specific example.

Let $x(t) = \begin{bmatrix} x_1(t) & x_2(t) & x_3(t) \end{bmatrix}'$

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \qquad A = \begin{bmatrix} 0 & 1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}.$$

a) Write explicitly, that is one by one, the three equations

$$E\dot{x} = Ax + Bu.$$

[2 marks]

b) Observe that one of the equations in part a) does not contain any time derivative. Solve this equation for the variable x_3 and use the solution to eliminate the variable x_3 from the differential equations and from the equation that defines the output signal. [2 marks]

c) Write the differential equations and the output equation computed in part b) in the form

$$\dot{\chi} = A_r\chi + B_r u, \qquad y = C_r\chi + D_r u,$$

with $\chi(t) \in \mathbb{R}^2$. Compute explicitly the matrices A_r, B_r, C_r and D_r. [4 marks]

d) The eigenvalues of the descriptor system are the solution of the equation $\det(\lambda E - A) = 0$. Compute the eigenvalues of the considered descriptor system and show that these coincide with the eigenvalues of the matrix A_r. [4 marks]

e) Show that the system determined in part c) is observable. The descriptor system is observable if

$$\operatorname{rank}\begin{bmatrix} \lambda E - A \\ C \end{bmatrix} = 3,$$

for all λ which are eigenvalues of the descriptor system. Show that the descriptor system is observable. [4 marks]

f) Show that the system determined in part c) is controllable. Show using the definition of controllability and the simplest possible argument that the descriptor system is also controllable. [4 marks]

4. Consider a nonlinear, continuous-time, system described by the equations

$$\dot{x} = f(x) = \begin{bmatrix} x_2^2 \\ x_3^2 \\ 0 \end{bmatrix},$$

with $x(t) = [x_1(t), x_2(t), x_3(t)]' \in \mathbb{R}^3$.

a) Compute all equilibrium points of the system. [2 marks]

b) Compute the linearization of the system around the equilibrium points determined in part a) and write explicitly the matrix A of the linearized system. (Hint: the matrix A is the same at all equilibrium points.) [4 marks]

c) Study the stability properties of the linearized system. [2 marks]

d) Solve the differential equations of the nonlinear system to determine $x(t)$ as a function of $x(0)$, hence argue that all equilibrium points of the system are unstable. [6 marks]

e) Consider the output equation $y = x_1$. Show that

$$\frac{d^4 y}{dt^4} = 0,$$

hence argue that the nonlinear system with output can be rewritten as a linear system with equations

$$\dot{x}_e = A_e x_e, \qquad y = C_e x_e,$$

with $x_e(t) \in \mathbb{R}^4$. Write explicitly the matrices A_e and C_e. [6 marks]